



Bond Futures

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What are Bond Futures?

- A standardized contract for the sale or purchase of a fixed amount of a government/ treasury bond at a future specified time
- Traded on exchanges (e.g. CME, ASX) for different currencies (e.g. USD, AUD)
- Example: A March 2025 futures contract on 10-year US Treasury bond
- Uses of Bond Futures:
 - Speculate on future bond prices/ interest rates
 - Hedge against movements in future bond prices/ interest rates

Bond Terminology

- **Maturity:** date on which the final payment is due
- **Face Value/ Principal:** amount to be repaid upon bond maturity
- **Coupon:** periodic interest payment received by the bondholder
- **Accrued Interest:** interest accrued since last coupon payment
- **Clean and Dirty Prices:** price of bond without accrued interest is clean price; dirty price includes accrued interest
 - Dirty Price = Clean Price + Accrued Interest
- **Yield-to-Maturity (YTM):** effective rate of return on the bond

$$B = \sum_{i=1}^N \frac{C_i}{(1 + Y/f)^{t_i}} + \frac{P}{(1 + Y/f)^{t_N}}$$

- Where B is the bond dirty price, t_i being the coupon (C_i) payment times, with t_N being the final coupon and principal (P) payment time, f is coupon frequency

Bond Futures Features

- Bond future final delivery and settlement mechanisms:
 - **Cheapest-To-Deliver (CTD) Bond:** In some markets (e.g. CME, Eurex), the bond futures is settled by the delivery of a bond from a basket. The future seller has the option to choose the bond and hence would seek to deliver the cheapest-to-deliver bond
 - **Average Cash-Settled:** In some markets (ASX), the bond future is cash-settled based on the average yields of the bonds in a basket.
 - **Single Bond:** In some other markets (e.g. Sweden, South Africa, Mexico), the bond futures refer to a single pre-determined bond

Delivery	Settlement	Example
CTD	Physical	Eurex, CME, TSE
CTD	Cash	Japan Mini 10Y
Pre-determined bond	Physical	Sweden, Mexico, South Africa
Averaging	Cash	Korea, Australia, New Zealand

Bond Futures Features

- Quotation Conventions:
 - Price quoted directly e.g. CME, Eurex, CFFEX.
 - Yield is quoted (e.g. Sweden) or $(1 - \text{Yield}) \times 100$ is quoted (e.g. Australia)
 - There is a deterministic arithmetic formula to convert from yield to price defined by the exchange.
- Conversion Factors for CTD Price Quotations:
 - Final delivery price = Settlement Price $\times$ Conversion factor + Accrued interest
 - The conversion factor for a bond is equal to the value of the bond (\$1 principal) assuming a fixed yield-to-maturity (e.g. 6% for US Treasury bonds)
 - For USD treasury bonds where the coupon exceeds 6%, the conversion factor is greater than 1, reflecting the higher relative value of the security. Conversely, if the coupon falls below 6%, the conversion factor is less than 1, indicating a comparatively lower value for the security.
 - The purpose of conversion factors is to put all bonds on a roughly equal footing irrespective of their coupon rate and their maturity.
 - For the same yield, bonds with a lower coupon will have a lower price than bonds with a higher coupons. Therefore, without the presence of a conversion factor, the bonds bearing the lowest coupons tend to be the cheapest.

Futures on US Treasury Bonds

- Contract Unit: \$100,000
- Quoted Price based on principal of \$100
 - E.g., 101.25, 99.9375
- Quarterly Contracts (March, June, Sep, Dec)
- Typically, the front 3 quarterly contracts are traded, and majority of positions are on the front contract:
- Tick size (i.e., minimum price fluctuation): 0.03125 (=1/32)
 - Tick value = $0.03125 \times \$1,000 = \31.25
- Delivery on any business day in the contract month
- Conversion Factor based on 6% yield
- Example basket: U.S. Treasury bonds with remaining maturity of 15-25 years

Bond Futures Cashflows

- Daily settlements are in cash
 - Day 1: the cash-settlement of the difference between today's end-of-day price and the trade price.
 - Days 2 to the expiry date: the cash-settlement of the difference between today's price and the previous day's price.
 - E.g., if the future price changes from 100.25 to 100.5 from one day to next, the settlement amount for one contract of \$100,000 is $\$0.25 \times 1000 = \250
- Final settlement date: the delivery (either cash or physical) of the bond against an amount based on the following:

Last trading day future's price (F)	100.75
Clean price of CTD bond on delivery date (P)	96.0
Conversion factor (CF)	0.95
P&L at delivery = $(P - F \times CF) \times \1000	\$287.5

Bond Forward

- Obligation for one party to transfer a specific bond to the other, at a specific settlement date
 - In the cash-settled variant, to transfer the market value of the bond at that date.
- The contract typically specifies a predetermined 'strike' price to be paid by the party receiving the bond at the time of bond settlement
- The bond future contract is economically similar to a bond forward contract but the

Bond Forward	Bond Future
Over the counter (OTC)	Exchange traded
Per-determined bond	CTD Option for future seller in some markets
Settled once	Settled daily
Fixed delivery date	Can be delivered any day in delivery month

Bond Forward Valuation

Stylized Example:

- Bond Repurchase Agreement (Repo): Bond used as collateral for a loan
- Prevailing repo rate for a security determines the cost of funding a temporary holding of a particular asset, and hence its fair forward price
- Assume that at each time t a “term repo” market exists in which it is possible to use bond as collateral for a loan until a later time T . Also assume no coupon between 0 and T .

Replication and No-Arbitrage Argument:

At time 0:

- Buy the bond in the market for price B ;
- Borrow amount B in the repo market until T using the bond as collateral $\Rightarrow$ Net cash = 0.

At time T :

- Repay $B(1+RT)$ to your repo counterparty (R being the repo rate);
- Recover the bond from your repo counterparty.

Hence, fair par forward price = $B(1+RT)$

Bond Forward Valuation

Adjusting for coupons:

- Suppose the bond pays coupons between 0 and T
- The bond owner is entitled to these coupons
- Hence, the cost of holding the bond from 0 to T needs to be reduced by the reinvested coupons amounts
- Fair Bond Forward Price = $B(1+RT) - \sum_i C_i(1+R_i(T-t_i))$
 - Where C_i is the i -th coupon paid at t_i between 0 and T and is assumed to be re-invested in repo market at rate R_i

Further Readings

- Options, Futures, and Other Derivatives by John C. Hull.
- US Treasury Bond Futures on CME Exchange website: <https://www.cmegroup.com/trading/interest-rates/us-treasury.html>
- Bond Futures on ASX Exchange website: <https://www.asx.com.au/markets/trade-our-derivatives-market/overview/interest-rate-derivatives/bond-derivatives>

Thank you

